Supplementary Material

This supplementary document provides additional materials referenced in the main paper, including:

(i) EEG topographic maps under alternative event-window lengths

(ii) Detailed frequency-band and regional summaries for the six emotions

(iii) Statistical analyses of potential order effects under the fixed emotional sequence

(iv) Dataset distribution summaries

S1 Different Time Windows for Cluster-Based Permutation Testing

To further validate the choice of the 4-s event-centered window in the main paper, we repeated the group-level cluster-based permutation analysis using adjacent 3-s and 5-s windows. Specifically, the 3-s window was defined as $[t_{event} - 2 \text{ s}, t_{event} + 1 \text{ s}]$, and the 5-s window was defined as $[t_{event} - 2 \text{ s}, t_{event} + 3 \text{ s}]$.

As in the main paper, Power Spectral Density (PSD) changes relative to baseline were computed across five standard frequency bands (δ, θ, α, β, γ), and statistical significance was assessed using cluster-based permutation testing. Channels belonging to statistically significant clusters are marked with white circles. Red/blue indicates PSD increase/decrease relative to baseline, and greyed-out maps indicate that no significant clusters were found.

S2 Detailed EEG Frequency-Band and Regional Summaries

Table S1 provides detailed summaries of the dominant frequency-band modulations and corresponding brain regions for the six target emotions. This table complements Fig. 17 in the main paper and is referenced there as Supplementary Table S1.

Table S1: Detailed EEG neurophysiological signatures of six basic emotions: dominant frequency-band modulations and regional patterns.

Emotion	Key Frequency Bands	Key Brain Regions
Happiness	α ($\downarrow$ 67.8%)	Frontal-parietal
	γ ($\uparrow$ 28.8%), β ($\uparrow$ 10.2%)	Temporal-frontal
Surprise	α ($\downarrow$ 35.6%)	Parietal-occipital
	β ($\downarrow$ 35.6%), γ ($\downarrow$ 27.1%)	Temporal-parietal
Disgust	α ($\downarrow$ 25.4%)	Frontal-central-parietal
	θ ($\downarrow$ 10.2%)	Frontal
Anger	α ($\downarrow$ 45.8%)	Frontal-temporal-central-parietal
	β ($\downarrow$ 15.3%)	Temporal-central-parietal
Fear	α ($\downarrow$ 91.5%)	Frontal-temporal-central-parietal
	β ($\downarrow$ 30.5%)	Frontal-central-parietal
	θ ($\downarrow$ 20.3%)	Central-parietal
Sadness	α ($\downarrow$ 81.4%)	Frontal-temporal-parietal-occipital
	β ($\downarrow$ 39.0%)	Frontal-temporal-parietal
	γ ($\downarrow$ 16.9%)	Parietal-occipital

Legend: $\uparrow$ Power increase (synchronization); $\downarrow$ Power decrease (desynchronization).

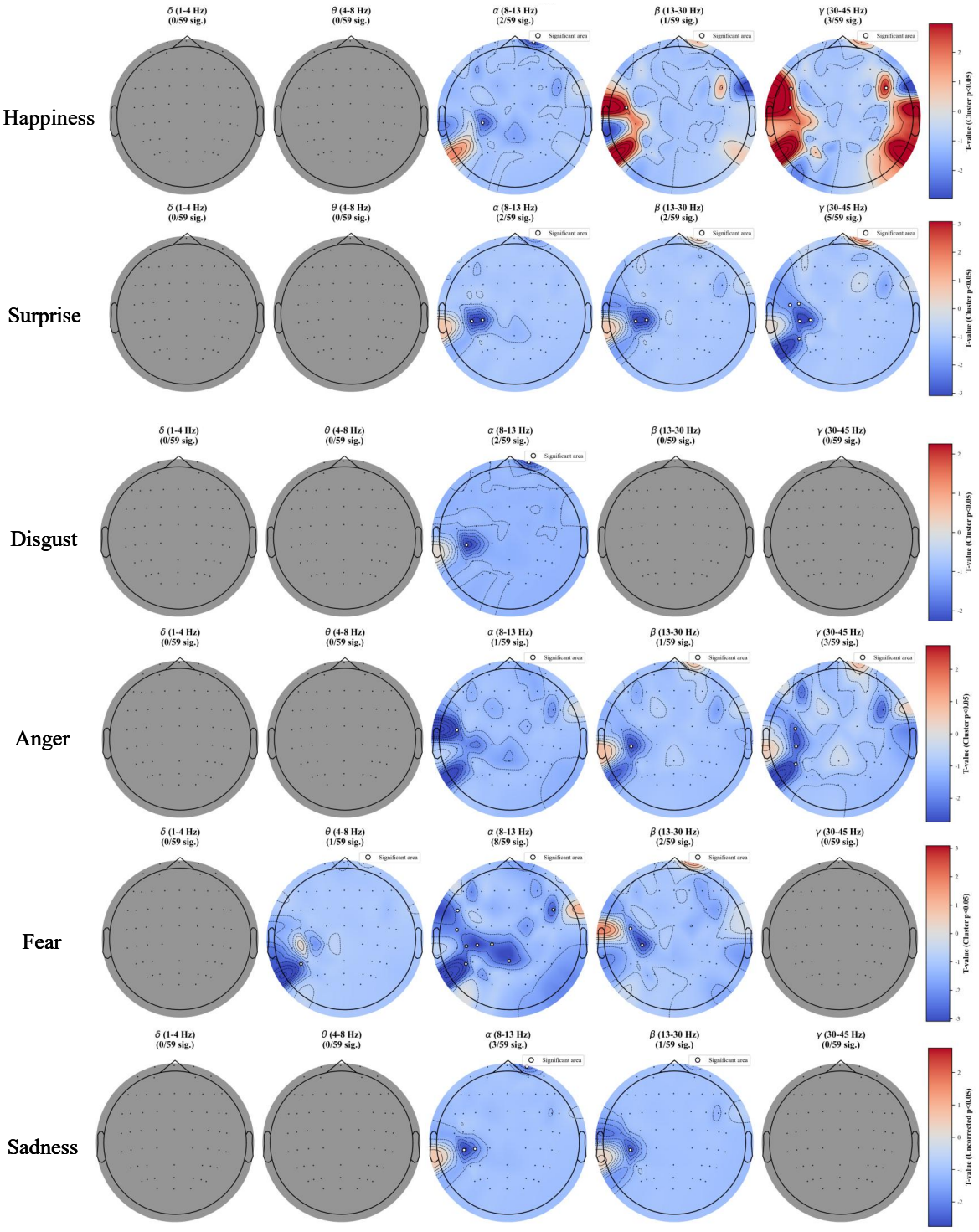

Figure S1: Event-related EEG PSD change T-maps (3-s window) for six emotions across five frequency bands. White circles indicate channels in significant clusters (cluster permutation test, $p < 0.05$). Red/blue indicates PSD increase/decrease relative to baseline. Greyed-out maps indicate no significant clusters were found.

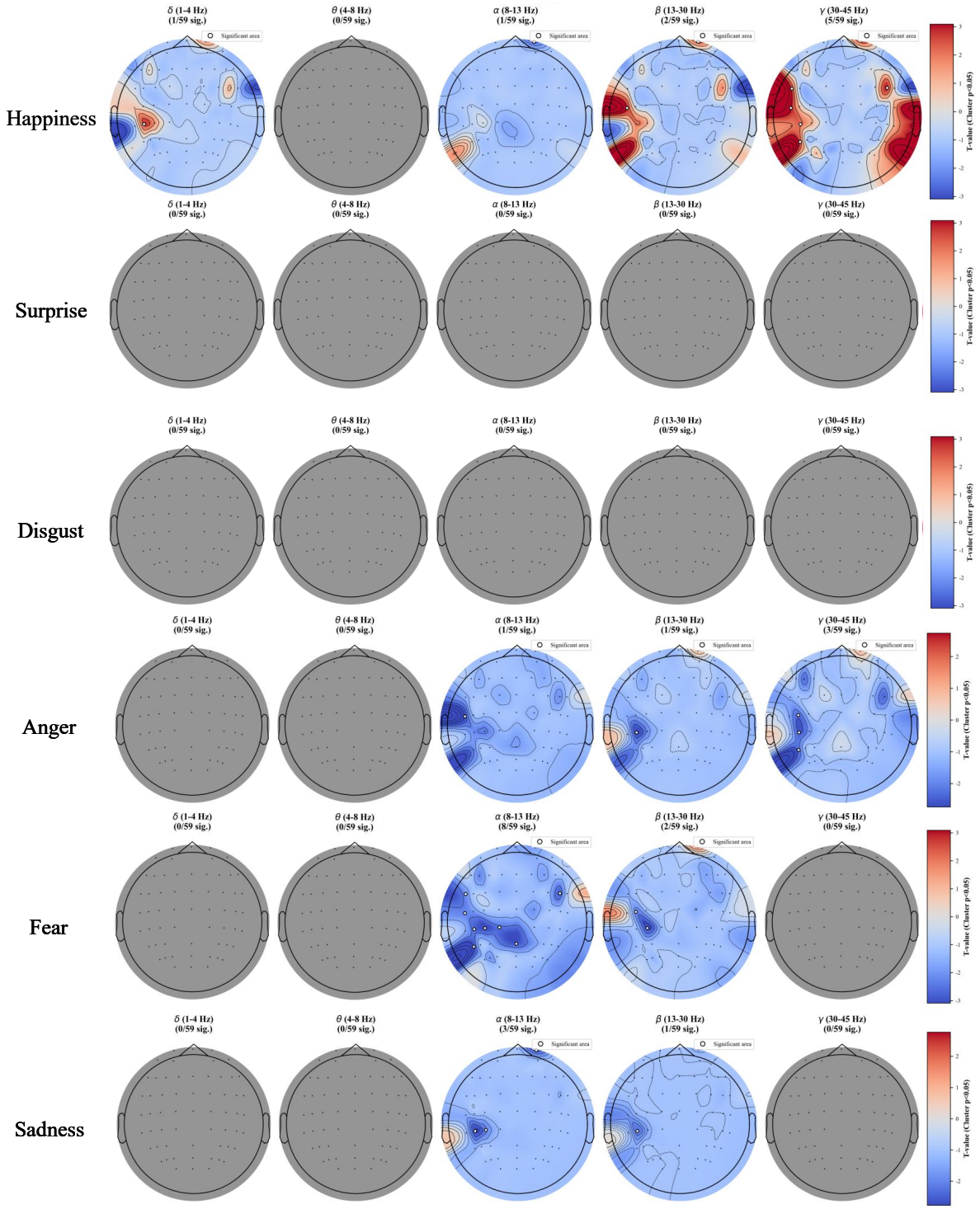

Figure S2: Event-related EEG PSD change T-maps (5-s window) for six emotions across five frequency bands. White circles indicate channels in significant clusters (cluster permutation test, $p < 0.05$). Red/blue indicates PSD increase/decrease relative to baseline. Greyed-out maps indicate no significant clusters were found.

S3 Analysis of Potential Order Effects

Because the main experiment used a fixed emotional sequence within each session, we conducted additional analyses to examine whether potential order effects systematically influenced the results.

We compared key dependent variables between the first emotion-cycle block (Block 1, trials 1–6) and the final emotion-cycle block (Block 5, trials 25–30). First, paired-samples t-tests were used to compare (i) baseline skin conductance level (SCL) and (ii) annotation count per trial. Second, a 2 (Time: Block 1 vs. Block 5) $\times$ 6 (Emotion) repeated-measures ANOVA was used to test whether subjective intensity ratings changed over time or showed emotion-specific habituation.

As shown in Tables S2 and S3, no significant main effect of Time or Time $\times$ Emotion interaction was observed. Likewise, no significant differences were found in baseline SCL or annotation count between Block 1 and Block 5. These results suggest that the fixed emotional sequence did not introduce significant systematic bias in baseline physiological arousal, annotation behavior, or subjective intensity ratings.

Table S2: Paired t-test results for physiological and behavioral variables (Block 1 vs. Block 5).

Variable	**Block 1 (M ± SD)**	**Block 5 (M ± SD)**	**Statistics**
Baseline SCL (μS)	2.32 ± 1.10	2.28 ± 1.00	$t(34) = 0.72,\ p = .47$
Annotation Count (per trial)	1.85 ± 0.65	1.81 ± 0.62	$t(34) = 0.41,\ p = .68$

Table S3: Repeated-measures ANOVA results for subjective intensity ratings (Block 1 vs. Block 5).

Effect	**df**	**F-value**	**p-value**	$\boldsymbol{\eta_p^2}$
Main Effect of Time	1, 34	0.58	.45	0.02
Time $\times$ Emotion Interaction	5, 170	0.89	.49	0.03

S4 Optional Dataset Distribution Summaries

Table S4: Per-emotion event distribution in FIRMED.

Emotion	**# Trials**	**# Events**	**Avg. Events/Trial**
Happiness	525	2880	5.49
Fear	525	2760	5.26
Surprise	525	2510	4.78
Disgust	525	2270	4.32
Sadness	525	2110	4.02
Anger	525	1960	3.73
Total	3150	14490	4.60